Management of Phenotypic, Genotypic and Environmental Data in Plant Breeding Projects using Frictionless Validation of Excel Templates

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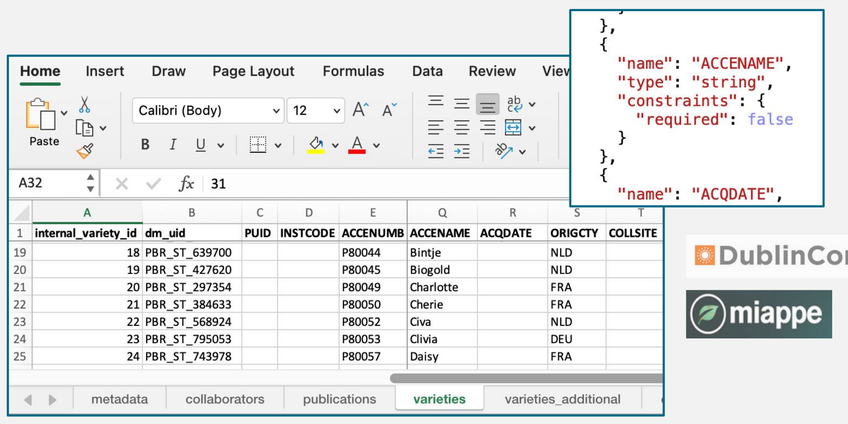
Effective management and analysis of genotypic, phenotypic, and environmental data generated by plant breeding initiatives are crucial for informed decision-making. However, many existing repository solutions may not meet the specific project requirements, forcing individuals to rely on Excel files for data storage, despite Excel not being an ideal option. Using such data files may lead to errors and complicated processing. To prevent such issues, we have developed an approach that utilizes Excel templates and a robust validation procedure to store project data.

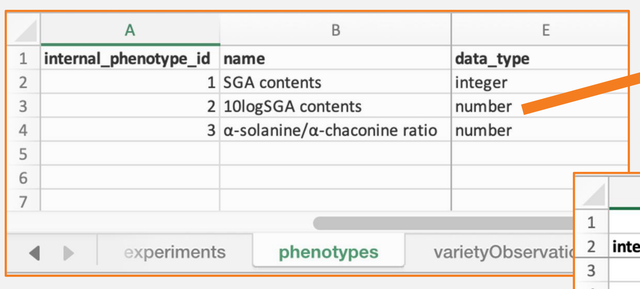
Project data stored in Excel

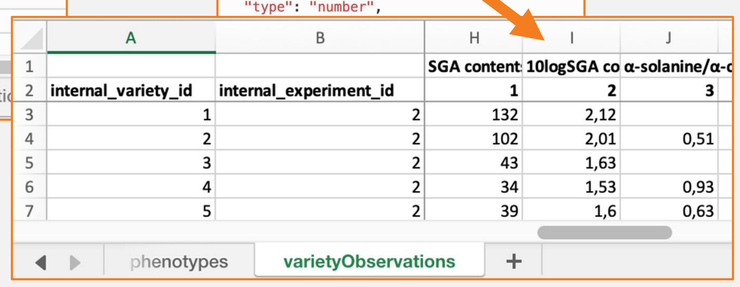


DublinCore

miappe

Schema for observation sheet is created **dynamically**, based on the defined phenotypes from the phenotype sheet





FRICTIONLESS DATA

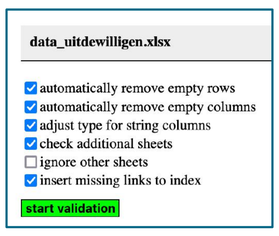
- Separate Excel file is used for each project
- Required and optional sheets
- For each sheet, a Frictionless schema is defined
- Structure and schemas based on existing standards: DublinCore, MCPD, MIAPPE
- Additional sheets can be included

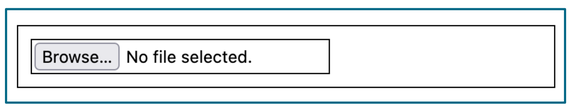
validation

After successful validation, the Excel file along with a metadata file containing the schemas can be conveniently stored as a comprehensive **Data Package**.

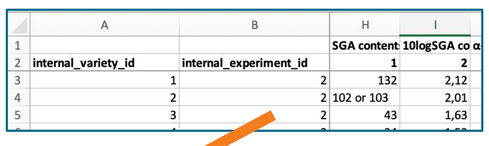
Validator

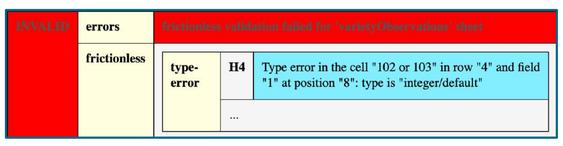
We provide a versatile **validation tool** that enables users to check Excel files against the schemas. This tool can be utilized conveniently through both a command line interface and a user-friendly **web interface**.

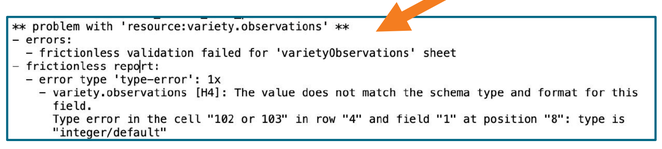




Detailed reports are automatically generated in case of errors, providing comprehensive insights for efficient troubleshooting.

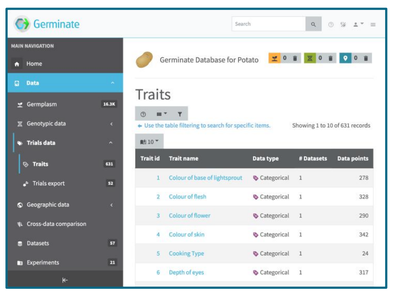






Process

An import scripts can be used to map values from validated and linked Excel files to an existing repository system, such as Germinate. Once imported, Germinate provides a convenient BrAPI endpoint for accessing and analysing the data.



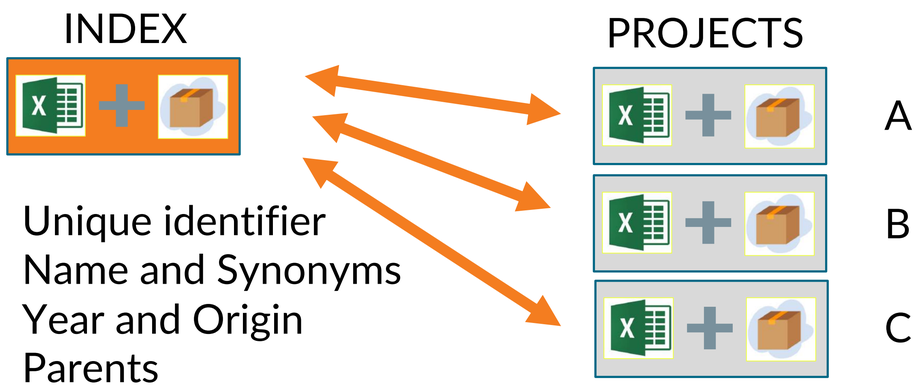
BrAPI

python

To facilitate analysis, standardized pipelines are implemented by leveraging two approaches: utilizing the BrAPI endpoint or working directly with validated Data Packages.

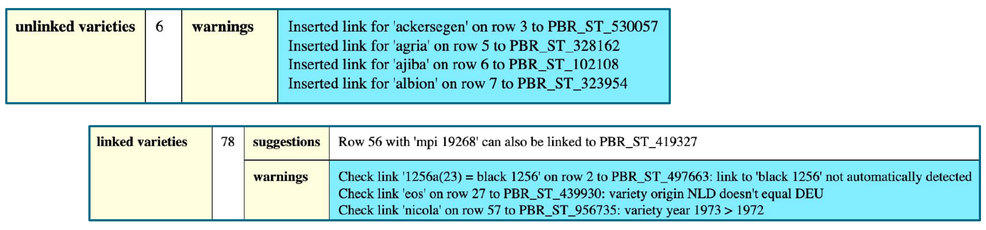
Index

A standardized and also frictionless validated Excel file with passport data and a **unique identifier** is used as an index of known accessions



- Unique identifier
- Name and Synonyms
- Year and Origin
- Parents

Accessions within a project can be **linked to the index** by including an identifier. This process has been largely automated and linking is tested in the validation tools.



Overall, our approach provides a straightforward yet effective way of managing data from different plant breeding experiments and projects while adhering to the FAIR principles.

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AGENT

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